

Virtual Workstations in Creative Arts Education: Evaluating Cloud-Based Delivery at Scale

Brad Nisbet*
Faculty of Fine Arts and Music
University of Melbourne
Melbourne, Victoria, Australia
brad.nisbet@unimelb.edu.au

Phillip Wilkinson
Victorian College of the Arts
University of Melbourne
Melbourne, Victoria, Australia
phillip.wilkinson@unimelb.edu.au

Bonnie Van De Ven
Victorian College of the Arts
University of Melbourne
Melbourne, Victoria, Australia
bonnie.vandeven@unimelb.edu.au

Abstract

This pilot study evaluates cloud-hosted virtual workstations delivered via Amazon Web Services (AWS) for real-time virtual production education using Unreal Engine, aiming to enhance accessibility for remote, regional, disabled, and socio-economically disadvantaged students. Conducted over two days through four structured sessions emulating authentic project-based learning, the research employed mixed-method data collection (including quantitative surveys and qualitative feedback from educators and students) to assess technical feasibility, user experience, and pedagogical outcomes.

Results indicated rapid participant adaptation, with initial access friction significantly decreasing and system usability metrics markedly improving over the course duration. Notably, latency experiences varied independently of internet speed, highlighting user interface and workload adaptation as critical factors influencing perceived performance. Educator reflections emphasized the necessity of targeted support ratios, ergonomic considerations, and adaptive pedagogical strategies tailored to virtual environments.

Findings demonstrate that cloud-based virtual workstations effectively overcome traditional barriers associated with on-campus computing resources, provided that pedagogical design prioritizes comprehensive user support, ergonomic awareness, and cognitive load management. This scalable approach offers promising implications for equitable, inclusive creative technology education.

CCS Concepts

• **Human-centered computing** → Human computer interaction (HCI); Empirical studies in HCI; • **Applied computing** → Education; Distance learning; Education; Interactive learning environments.

ACM Reference Format:

Brad Nisbet, Phillip Wilkinson, and Bonnie Van De Ven. 2025. Virtual Workstations in Creative Arts Education: Evaluating Cloud-Based Delivery at Scale. In *SIGGRAPH Asia 2025 Educator's Forum (SA Educator's Forum '25)*, December 15–18, 2025, Hong Kong, Hong Kong. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3757372.3771847>

*Correspondence author

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
SA Educator's Forum '25, Hong Kong, Hong Kong
© 2025 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2132-8/2025/12
<https://doi.org/10.1145/3757372.3771847>

1 Introduction

Creative arts education increasingly demands the integration of high-performance graphical computing hardware to support real-time 3D graphics software environments (Epic Games. , 2025). Yet access to such environments is often restricted to specialized computing labs located on campus, limiting participation for remote, regional, disabled, and socio-economically disadvantaged students.

This pilot project explored the feasibility and educational impact of using cloud-based virtual workstations hosted by Amazon Web Services (AWS) to deliver real-time virtual production and Unreal Engine teaching through a custom-built remote learning environment. We present the results from our 4-session pilot (totalling 12 teaching hours over 2 days) using a mixed-method data collection, combining quantitative survey results and qualitative reflections from educators and students. Our contribution includes a review of system performance, pedagogical observations, and practical recommendations for future large-scale deployments.

2 Methodology

2.1 Study Design and Purpose

This pilot study aimed to evaluate the feasibility, accessibility, and pedagogical viability of cloud-hosted virtual workstations for delivering advanced digital arts education. The technology under evaluation (GPU-enabled AWS EC2 instances pre-configured with industry-standard software) was deployed as an alternative to a traditional on-campus high-performance computing lab.

The study adopted a mixed-methods research design to generate a holistic understanding of the user perceptions of both the technical performance and pedagogical affordances of virtual workstations in a remote creative learning context (Creswell and Clarke, 2018). Quantitative data was gathered through structured surveys capturing user-reported metrics such as latency, task efficiency, technical difficulties, and overall satisfaction.

Both numerical data and qualitative insights were obtained through anonymous participant feedback and structured observational reports from academic teaching staff. The addition of open-ended questions was also designed to generate genuine accounts that would provide a richer narrative of user experience and any barriers encountered throughout the pilot (Komildjanovna, 2024). Meanwhile, the inclusion of educator reflections was intended to consider critical insights into facilitation workflows, scalability concerns, pedagogical considerations, and virtual classroom dynamics.

The study design integrates elements of:

- A usability study

- evaluating *access*, latency, and performance under real conditions
- Observational field research
- documenting behaviour, adaptation, and instructional response
- An educational field trial
- simulating *course* delivery and learner outcomes in an authentic but experimental setting

By triangulating multiple forms of data across sessions and user roles, the study captures both micro-level user experience and macro-level educational system dynamics, providing a robust foundation for future implementation and scaling (Creswell and Creswell, 2022).

It is important to acknowledge that this study's duration (four sessions over two days) and sample size ($n=19$) reflect a pilot-scale implementation. As such, findings regarding pedagogical impact and scalability are exploratory rather than definitive.

2.2 Participant Recruitment and Consent

Participants were recruited via direct invitations sent to alumni mailing lists, directing them to complete an expression of interest (EOI) webform. Those who submitted both the EOI and consent form were invited to an online onboarding session outlining the study details and participation expectations. All participants were adult volunteers with an interest in Unreal Engine or Creative Technology. Of 161 invited, 24 completed consent forms by the deadline; 5 later withdrew, resulting in 19 participants.

2.3 Procedure and Session Structure

The sessions were structured as a simulation of a future micro-credential course offering, "*Telling Stories with Unreal Engine*", designed to emulate authentic creative learning environments and project-based workflows.

The study ran over four sessions across two days:

Day 1 AM: Introduction to Unreal Engine, navigation, asset handling, and interface.

Day 1 PM: Introduction to Animation Workflows in Unreal Engine.

Day 2 AM: Animation and Sequencer in Unreal Engine.

Day 2 PM: Workflow tasks, real-time lighting, camera work, and rendering.

Each session was 3 hours, with guided instruction from a lecturer and support from a tutor. The online instruction was delivered via the Zoom platform while students accessed their virtual workstations via the Amazon DCV client software (a low latency remote virtual desktop application) (Amazon Web Services, 2025). Participants completed short, anonymized surveys after each session, assessing technical experience, access, latency, efficiency, and general satisfaction.

2.4 Data Analysis

To analyse the qualitative data, I employed a reflexive thematic analysis grounded in Braun and Clarke's (2006) inductive coding methodology. This approach was chosen because of its flexibility and capacity to reveal nuanced patterns in participant experience without imposing a predetermined framework. The lead researcher

had prior experience using this method and adapted it to suit the study's aims and time constraints by integrating support from a custom large language model (LLM) based on LLAMA 3.3 (Ollama, 2025). The LLM was used to assist with rapid text sorting, code clustering, and initial thematic suggestion.

To ensure rigor in LLM-assisted thematic analysis, outputs from the model were cross-validated by the lead researcher, who has prior experience in reflexive qualitative coding. The LLM (LLAMA 3.3) was not used to replace interpretive judgment but to expedite clustering and surface thematic candidates. Ambiguous or low-confidence codes were manually reviewed, and potential bias from training data was mitigated by re-contextualizing outputs against original source quotes. This hybrid method offers both efficiency and human oversight, though further methodological development is needed to ensure reproducibility across studies using LLM-supported qualitative analysis.

Sentiment and emotional tone detection were conducted using large language model (LLM) assistance, drawing on capabilities comparable to those offered by tools such as VADER (Hutto and Gilbert, 2014) and IBM Watson Natural Language Understanding (IBM., 2025), though implemented within a customised GPT-LLM environment. These tools were used to extract polarity (positive/negative/neutral tone) and infer psychological states (e.g., confusion, confidence, frustration) serving as a heuristic layer to guide human thematic coding.

2.5 Ethical Oversight

This research was approved by the University of Melbourne Human Research Ethics Committee. All participant identities were anonymized, and no personally identifiable data was collected at any stage. Surveys were administered via university-managed systems, with data stored securely in accordance with institutional protocols. Informed consent was obtained using plain language statements and protocols aligned with the National Statement on Ethical Conduct in Human Research (NHMRC, 2023).

3 Results

3.1 Student Experience Trends

Across the four-session pilot, key usability and engagement metrics improved markedly, reflecting a steep learning curve that began to flatten within the first day of use.

3.1.1 Access & Onboarding. On Day 1 AM, only 47% of participants found the virtual workstation "extremely easy" to access. By the Day 1 PM session this jumped to 85%, indicating rapid user adaptation and the effectiveness of support interventions. Access ease remained high on Day 2, with 90% of participants in the AM session selecting "extremely easy."

This sharp increase supports the assertion that onboarding difficulties were front-loaded, and once resolved, user confidence and fluency improved quickly.

3.1.2 Lag & Latency. Lag and latency scores varied significantly by session. The most common experience in Day 1 AM was "occasionally" (46.6%) and "rarely" (26.6%). By Day 2 PM, 43% of participants reported "rarely" experiencing lag, and only 14% reported frequent disruptions.

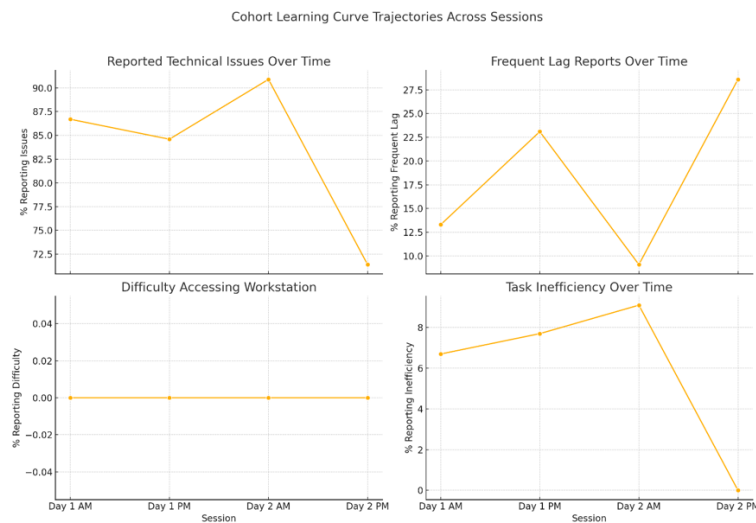


Figure 1: A graphical plot of the cohort learning curve trajectories across the four sessions.

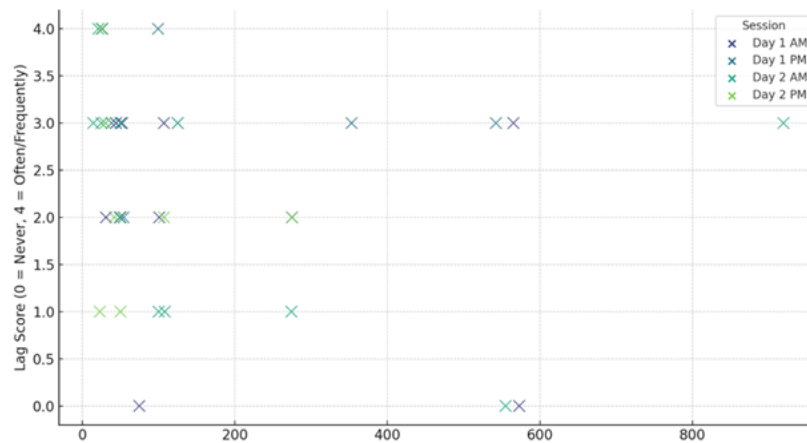


Figure 2: A scattered plot graph showing clusters of reported network speeds and instances of perceptible lag.

Key insights:

- The percentage of participants reporting frequent or consistent lag dropped by over 50% from Day 1 AM to Day 2 AM.
- User perception of lag was not strictly correlated with network speed, reinforcing that latency perception is as much about UX design and workload adaptation as it is about raw bandwidth.

3.2 Correlation and Contributing Factors

3.2.1 Connection Speed vs. Lag. Internet connection speed did not strongly predict user-reported lag. Participants with 30–50 Mbps reported smooth performance, while others with >500 Mbps still experienced jitter.

Network latency (ping) data collection was intended, however the submitted data from some participants (self-reported) indicated ping times that would be inconsistent with the laws of physics

given geographical constraints. This was determined to be due to a failure in the tool provided. As a result, the researchers decided not to include the data in the analyses.

Correlation analysis revealed a Pearson correlation ($r = -0.13$) between download speed and lag score of 'weak at best'.

Speed tiers:

- High (200+ Mbps): Average lag score = 1.9
- Medium (50–199 Mbps): Avg. lag = 2.2
- Low (<50 Mbps): Avg. lag = 2.93

Under 50 Mbps proved to be a risk threshold for stability, though not a guarantee of disruption.

3.2.2 Lag vs. Satisfaction & Efficiency. Satisfaction strongly negatively correlated with lag ($r = -0.64$), the most significant relationship observed. Meanwhile, lag vs. efficiency showed a weaker negative correlation ($r = -0.13$) which suggests some users were

Participant Geographic Region

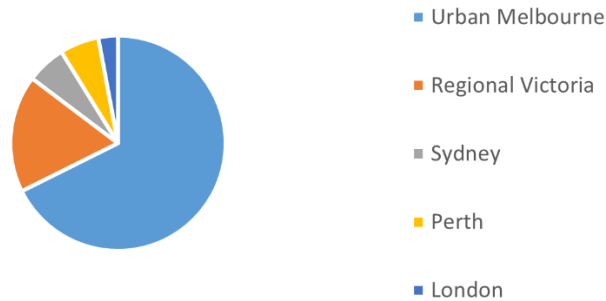


Figure 3: A pie chart showing participant geographical distribution.

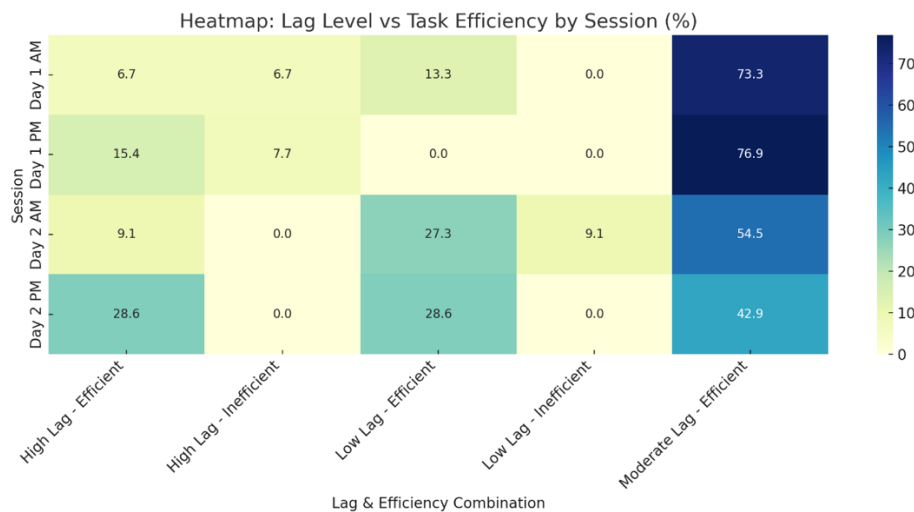


Figure 4: A heatmap showing the correlation between reported levels of "lag" and reported task efficiency.

still able to complete tasks effectively despite interruptions (likely due to task design, pacing, or user resilience).

3.3 Lag-Efficiency Interaction

To better understand the nuanced relationship between system performance and user outcomes, we conducted a cross-tab analysis of reported lag against perceived task efficiency across sessions which shows:

- In Day 1 AM, 73.3% of participants reported moderate lag but still completed tasks efficiently, this suggests a baseline tolerance or novelty-driven motivation.
- By Day 2 AM, 27.3% of users reported low lag and high efficiency (an ideal interaction).
- In Day 2 PM, results polarized: some users experienced high lag and still worked efficiently (28.6%), while others with low lag reported inefficiency. This may suggest fatigue, task complexity, or UI friction as a factor at this point in the study.

These shifting patterns suggest that while technical performance plays a role, actors such as user resilience, cognitive load, and instructional pacing are equally critical in participant satisfaction of the overall learning experience.

3.4 Technical Issues

To identify where technical friction emerged most prominently, participant-reported issues were classified according to their likely root cause:

- **Unreal Engine:** 11+ reports (including save/load errors, material glitches, timeline bugs, and crashes).
- **AWS Workstation:** 6+ reports (including full-screen lock-outs, screen scaling problems, and UI freezing).
- **Internet-related:** 10+ (disconnections, Zoom drops, and latency surges)

Despite this, the volume and severity of issues declined sharply after Session 1. By Day 2 PM, no critical failures were reported,

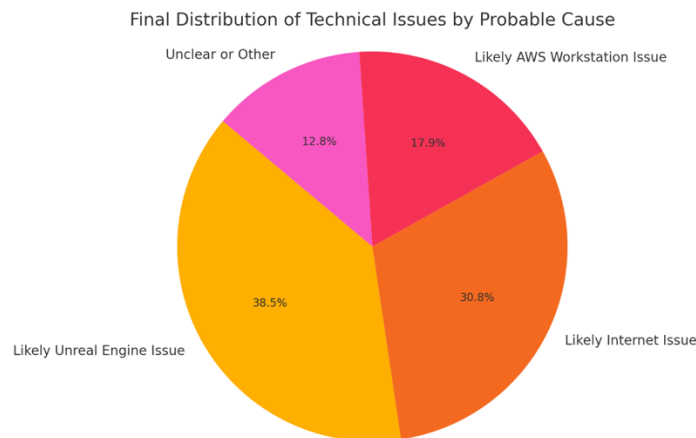


Figure 5: A pie chart showing the probable causes of technical issues reported by participants.

suggesting that both users and systems had adapted (or stabilised) across the duration of the pilot.

3.5 Lecturer Observations

To contextualize student experience data, educator reflections were thematically coded across 35 distinct entries spanning both teaching sessions and post-session feedback. These reflections provided critical insight into how instructors navigated the unique demands of facilitating real-time technical learning in a virtualized environment.

- **Technical Setup Friction:** Unreal Engine resource demands, Mac vs. PC inconsistencies, and interface scaling were consistent pain points.
- **Teaching Limitations:** Zoom limited non-verbal feedback; instructors struggled to read engagement or track progress mid-lesson.
- **Infrastructure Constraints:** Full-screen rendering bugs and dual-screen navigation issues affected multitasking and user autonomy.
- **Human Factors:** Ergonomic strain and cognitive fatigue emerged as notable factors in longer sessions.
- **Scalability Bottlenecks:** Instructors needed to support up to 3 students simultaneously, recommendation for a 1:5 support ratio, especially in Session 1.
- **Positive Feedback:** Virtual workstation remote support capabilities, scalability for rural deployment, and strong student engagement were cited repeatedly as major strengths.

These insights not only validated many of the trends observed in participant data but also exposed operational factors (such as pacing, ergonomics, and support load) that will be critical to address as this model scales beyond controlled pilot conditions.

4 Discussion

4.1 Learning Curves and System Adaptation

The pilot confirmed the presence of a clear user learning curve, both in terms of navigating the virtual workstation interface and

managing the demands of remote technical workflows. Metrics improved across every measured indicator between Session 1 and Session 4 with task efficiency rising from 71% to 92%, and technical issue frequency dropping by 64%. These trends, combined with rapid increases in reported ease-of-access, suggest that the primary friction points were front-loaded and not persistent once resolved.

However, the pacing and structure of the initial sessions required substantial facilitator support to mitigate student disruptions and prevent individuals from falling behind. This was particularly crucial during onboarding, as limited prior exposure to the software environment amplified the impact of minor technical issues across the class. Maintaining a 1:5 facilitator-to-student ratio (recommended by teaching staff) would minimize disruptions to the broader group, ensure learners remain engaged without being left behind, and significantly reduce the cognitive load on teaching staff, lowering the risk of fatigue and burnout during scaled implementations.

4.2 Latency, Satisfaction, and Misleading Metrics

One of the most interesting findings was the disconnect between network speed and latency and user experience. Despite some participants having high-speed connections (>500 Mbps), perceived lag still occurred whereas others with slower speeds (~30 Mbps) reported no issues. This reinforces the need to avoid relying on network speed alone as a proxy for readiness and instead consider a more nuanced mix of device specs, local connection type (Wi-Fi vs. Ethernet), and digital literacy.

Where correlation did emerge, it was impactful:

- Lag was the strongest predictor of user satisfaction ($r = -0.64$).
- Efficiency was more loosely related, with some users maintaining performance despite moderate lag, possibly due to familiarity with the software, stronger problem-solving skills, or a greater digital literacy.

These patterns suggest that technical friction, while influential, is only one part of the equation. Participant outcomes were also

shaped by a constellation of human factors, including digital literacy, resilience under pressure, comfort with ambiguity, and the clarity of instructional design. Some participants navigated moderate lag or momentary system disruptions without a noticeable drop in task efficiency, indicating that prior exposure to creative software, problem-solving mindset, and cognitive adaptability played a significant role in mitigating the impact of technical limitations. This highlights the need to consider both system design and learner readiness when deploying cloud-based educational platforms.

It should be noted that the study primarily relied on self-reported data for measures such as lag, efficiency, and satisfaction. While this provided valuable insight into user perceptions, future implementations will incorporate objective metrics (e.g., automated session logs, system telemetry, completion timestamps) to validate participant responses and identify latent usability issues. Integrating observational and log-based data will strengthen triangulation and help distinguish between subjective friction and verifiable performance limitations.

4.3 Pedagogical Design in Virtual Creative Spaces

While the technical architecture functioned reliably after initial setup, the pedagogical implications were more complex. Lecturers consistently reported challenges in reading student understanding, managing multiple channels of communication, and responding in real time (all consequences of teaching without sharing the same physical location as the students).

Moreover, the virtual environment inhibited peer-to-peer interaction, which is known to boost learning in collaborative creative contexts. Without hallway conversations, peer-to-peer learning, or lab-side problem-solving, much of the burden fell back on instructors. Even with jump-in access to student VMs, the experience lacked the ambient awareness of face-to-face teaching.

To mitigate these effects, future iterations should consider:

- Embedding structured collaborative tasks or peer roles within the curriculum
- Implementing voice or text-based breakout spaces for parallel collaboration
- Using more suitable tools beyond Zoom to support more nuanced instructional communication

While a hybrid approach (such as having some learners on-site in thin client PC labs connecting to their virtual machines) may solve this issue for some, further design that allows for mixed co-presence in both virtual and physical spaces will need to carefully consider factors for ensuring equitable experiences for the online and the on-site learners.

4.4 Infrastructure, Fatigue, and Human-Centred Design

Though rarely discussed in technical deployments, ergonomics and attention fatigue emerged as key concerns. Lecturers reported physical discomfort, screen management frustration, and cognitive overload (often exacerbated by the demands of simultaneous VM monitoring, chat handling, and lecture delivery). Students, too, were affected. The absence of physical cues or embodied interaction

meant longer sessions required intentional design around pacing, movement, and variation.

As virtual workstation delivery scales, attention must shift from proving technical feasibility to refining the human experience of virtual learning. This includes:

- Session structuring that prioritizes variation and recovery
- Environment-aware UI configurations (e.g., better multi-monitor support)
- Ergonomic considerations, including scheduled movement breaks and interface flexibility

4.5 Implications for Scalability and Equity

Perhaps the most significant takeaway from the pilot is that cloud-based virtual workstations can deliver powerful, industry-standard creative tools at scale (without the footprint, infrastructure overhead, and rigid capacity limits of traditional on-campus lab environments). This is a major enabler for equity, particularly for students in rural, remote, or socioeconomically disadvantaged settings.

However, this potential can only be realized if delivery models account for:

- The additional pedagogical overhead of remote learning
- The technical and emotional support structures needed to prevent early disengagement
- The holistic design of curriculum and session flow for virtual delivery

Without these considerations, virtual infrastructure risks replicating the same barriers it aims to dismantle, simply in a new modality.

While this study prioritised inclusion of diverse geographic regions and invited participants from underrepresented backgrounds, sample size constraints limited formal disaggregation by socioeconomic status, disability, or prior experience. Future studies with larger cohorts will incorporate stratified demographic analysis to more rigorously examine equity impacts and identify variation in experience across different learner groups.

5 Conclusion

This pilot study suggests that cloud-hosted virtual workstations are not only technically feasible but pedagogically viable for delivering advanced creative technology education at scale. Despite early-stage friction, participants quickly adapted to the virtual environment, with measurable improvements in task efficiency, user satisfaction, and system fluency over time.

The infrastructure proved stable across a range of internet speeds and geographic locations, suggesting that access to high-performance creative tools no longer needs to be tethered to physical high performance computing labs. This model eliminates many of the traditional spatial and logistical constraints of on-site delivery (including lab bookings, high-end hardware provisioning, and floorplan limitations) unlocking new opportunities for scalable, location-agnostic education.

While this study focused on technical and pedagogical feasibility, institutional implementation will also depend on cost-efficiency. Factors such as cloud compute instance costs, licensing, and staff-to-student support ratios must be considered in future economic modelling. A cost-benefit analysis comparing virtual workstation

models with traditional on-campus lab provisioning is a priority for subsequent research, particularly when planning for scale and sustained delivery across diverse learning contexts.

Yet this shift is not without trade-offs. The absence of embodied cues, the reliance on mediated communication, and the multitasking burden placed on instructors all reveal that virtual delivery requires more than just infrastructure, it demands a systemic redesign of learning workflows. Human factors, such as digital literacy, resilience, ergonomic design, and instructional pacing, remain critical determinants of success. Problem-based learning and assessment design, as well as a constructivist pedagogy are well-aligned to take advantage of these new tools (Hassan et al., 2022).

The pilot also reaffirms the necessity of robust onboarding, targeted facilitation ratios, and continuous adaptation of both tools and pedagogy to meet learners where they are; cognitively, physically, and geographically.

Looking forward, the success of this model hinges on how institutions choose to scale it, not just in size, but in quality, inclusivity, and responsiveness. Future iterations should explore integration with micro-credentials, cross-institutional collaborations, and pathways that extend this access to underrepresented learners, rural

students, and creative practitioners previously excluded by hardware and location constraints.

The future of creative education may not be bound to a place, but it must remain anchored in human-centered design, equitable access, and purposeful pedagogical intent.

References

- Amazon Web Services. (2025, April 30). Amazon DCV. Retrieved from Downloads: <https://www.amazondcv.com>
- Creswell, & Clarke. (2018). *Designing and Conducting Mixed Methods Research*. SAGE.
- Creswell, J. W., & Creswell, J. D. (2022). *Research Design: Qualitative, Quantitative, and Mixed Methods Approaches*. SAGE.
- Epic Games. (2025, April 30). Unreal Dev Forums. Retrieved from Hardware and Software Specifications : <https://dev.epicgames.com/documentation/en-us/unreal-engine/hardware-and-software-specifications-for-unreal-engine>
- Hassan, J., Devi, A., & Ray, B. (2022). Virtual Laboratories in Tertiary Education: Case Study Analysis by Learning Theories . *Educ. Sci.* 12(8).
- Hutto, C. J., & Gilbert, E. (2014). VADER: A Parsimonious Rule-based Model for Sentiment Analysis of Social Media Text. Georgia Institute of Technology.
- IBM. (2025, April 30). IBM Watson Natural Language Understanding . Retrieved from IBM - Watson: <https://www.ibm.com/products/natural-language-understanding>
- Komildjanovna, N. M. (2024). The Power Of Open-Ended Question And Its Crucial Role In Discovering Unanticipated Answers, Gathering Additional Details And Important Insights. *Web of Teachers: Inderscience Research*.
- Ollama. (2025, April 30). Ollama. Retrieved from Models: <https://ollama.com/library/llama3.3>